KronHiPPO-STGP: Online Bayesian Spatiotemporal Modeling with Evolving Memory

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Abstract

Spatiotemporal modeling often requires capturing both global trends and correlated uncertainty across space and time. In addition, since data arrive sequentially in typical applications, online modeling and prediction are required. We introduce KronHiPPO-STGP, an online Bayesian spatiotemporal model that jointly updates a parametric global trend (captured by linear regression) and a residual model by a Gaussian process (GP) with a separable spatiotemporal kernel. To maintain fixed memory usage during online inference, we use Online HiPPO Sparse Variational GP (Online HiPPO-SVGP) along the temporal axis, which updates the interdomain inducing variables by evolving an ODE-based recurrent memory. To address computational speed challenges, we solve the ODE evolution via Bessel function computations, and we leverage the Kronecker structure of the residual GP to avoid dense matrix inversion. These have reduced the computational complexity from $\mathcal{O}(M^6)$ to $\mathcal{O}(M^3)$, where M is the number of inducing variables along each spatial and temporal axis. Experiments on climate, public health and traffic modeling datasets show that KronHiPPO-STGP maintains memory costs independent of stream length and achieves accurate predictions with uncertainty quantification in online regression settings.

1 INTRODUCTION

Spatiotemporal data arise in real-world sequential records, such as climate observations (Muñoz-Sabater et al., 2021), public health monitoring data (Silk et al., 2023), and traffic records (Li et al., 2018). To model these data, a model needs to capture not only overall trends but also spatially and temporally correlated uncertainty. A standard approach to modeling spatiotemporal data is to combine a mean function, typically linear regression using spatial and temporal features, with a residual model, such as a Gaussian process (GP) with a separable kernel comprising spatial and temporal components (Cressie, 1993; Stein, 1999; Rasmussen and Williams, 2005; Flaxman et al., 2015). This decomposition provides a natural framework for capturing both global mean trends and correlated uncertainty across space and time.

In real-world applications, data arrive sequentially and the temporal horizon expands, so we need to update predictions in an online manner. However, repeatedly fitting the model, especially the residual GP part, using all accumulated data at each update requires a computational cost that grows with the temporal horizon (Bui et al., 2017). Therefore, a compressed representation is required to summarize the data streams (Csató and Opper, 2002; Titsias, 2009; Maddox et al., 2021).

To address this challenge, we construct temporal inducing variables for the residual GP based on Online HiPPO Sparse Variational GPs (Online HiPPO-SVGPs, OHSVGP; Chen et al., 2025), which use interdomain inducing variables (Lázaro-Gredilla and Figueiras-Vidal, 2009) defined on an orthogonal polynomial basis, with the projection given by solving the HiPPO ODE (Gu et al., 2020). This approach allows us to maintain a fixed-size representation of the temporal history, which is then updated by a joint online Bayesian inference method for the joint posterior of the linear trend and the residual GP.

Two further computational difficulties arise in this online setting.